Q1 - 2024 (04 Apr Shift 1)

If $\lim_{x \rightarrow 1} \frac{(5x+1)^{1/3}-(x+5)^{1/3}}{(2x+3)^{1/2}-(x+4)^{1/2}} = \frac{\mathrm{m}\sqrt{5}}{\mathrm{n}(2\mathrm{n})^{2/3}}$, where $\gcd(\mathrm{m}, \mathrm{n}) = 1$, then $8\ \mathrm{m} + 12\mathrm{n}$ is equal to______

Q2 - 2024 (04 Apr Shift 2)

Let $f(x) = \int_0^x \left(t + \sin\left(1 - e'\right)\right) dt, x \in \mathbb{R}$. Then, $\lim_{x \rightarrow 0} \frac{f(x)}{x^3}$ is equal to

(1) $-\frac{1}{6}$

(2) $\frac{2}{3}$

(3) $-\frac{2}{3}$

(4) $\frac{1}{6}$

Q3 - 2024 (05 Apr Shift 1)

Let f be a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $2f(2) + 3f(3)$ is equal to _______

Q4 - 2024 (05 Apr Shift 2)

Let $\mathrm{a} > 0$ be a root of the equation $2x^2 + x - 2 = 0$. If $\lim_{x \rightarrow \frac{1}{\mathrm{a}}} \frac{16\left(1 - \cos\left(2 + x - 2x^2\right)\right)}{(1 - \mathrm{a}x)^2} = \alpha + \beta\sqrt{17}$, where $\alpha, \beta \in Z$, then $\alpha + \beta$ is equal to_______

Q5 - 2024 (06 Apr Shift 1)

Let $f : (-\infty, \infty) - \{0\} \rightarrow \mathbb{R}$ be a differentiable function such that $f'(1) = \lim_{a \rightarrow \infty} a^2 f\left(\frac{1}{a}\right)$. Then $\lim_{a \rightarrow \infty} \frac{a(a+1)}{2} \tan^{-1}\left(\frac{1}{a}\right) + a^2 - 2\log_e a$ is equal to

(1) $\frac{3}{2} + \frac{\pi}{4}$

(2) $\frac{3}{4} + \frac{\pi}{8}$

(3) $\frac{3}{8} + \frac{\pi}{4}$

(4) $\frac{5}{2} + \frac{\pi}{8}$

Q6 - 2024 (06 Apr Shift 2)

$\lim_{n\to\infty}\frac{(1^2-1)(n-1)+(2^2-2)(n-2)+\cdots+((n-1)^2-(n-1))\cdot 1}{(1^3+2^3+\cdots\cdots+n^3)-(1^2+2^2+\cdots\cdots+n^2)}$ is equal to :

(1) $\frac{2}{3}$

(2) $\frac{1}{3}$

(3) $\frac{3}{4}$

(4) $\frac{1}{2}$

Q7 - 2024 (08 Apr Shift 1)

The value of $\lim_{x\to 0} 2\left(\frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}\ldots\ldots\sqrt[10]{\cos 10x}}{x^2}\right)$ is

Q8 - 2024 (08 Apr Shift 2)

If $\alpha=\lim_{x\to 0^+}\left(\frac{\mathrm{e}^{\sqrt{\tan x}}-\mathrm{e}^{\sqrt{x}}}{\sqrt{\tan x}-\sqrt{x}}\right)$ and $\beta=\lim_{x\to 0}(1+\sin x)^{\frac{1}{2}\cot x}$ are the roots of the quadratic equation $ax^2+bx-\sqrt{\mathrm{e}}=0$, then $12\log_{\mathrm{e}}(\mathrm{a}+\mathrm{b})$ is equal to__________

Q9 - 2024 (09 Apr Shift 1)

Let $\lim_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{(n^2+1)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8n}{(n^2+4)\sqrt{n^4+16}}+\ldots+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{(n^2+n^2)\sqrt{n^4+n^4}}\right)$ be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to ________

Q10 - 2024 (09 Apr Shift 2)

$\lim_{x\to\frac{\pi}{2}}\left(\frac{\int_{x^3}^{(\pi/2)^3}\left(\sin\left(2t^{1/3}\right)+\cos\left(t^{1/3}\right)\right)dt}{\left(x-\frac{\pi}{2}\right)^2}\right)$

is equal to

(1) $\frac{5\pi^2}{9}$

(2) $\frac{9\pi^2}{8}$

(3) $\frac{11\pi^2}{10}$

(4) $\frac{3\pi^2}{2}$

Q11 - 2024 (09 Apr Shift 2)

$\lim_{x\to 0} \frac{e-(1+2x)^{\frac{1}{2x}}}{x}$ is equal to

(1) 0

(2) $\frac{-2}{e}$

(3) e

(4) $e - e^2$

Answer Key

Q1 (100) **Q2** (1) **Q3** (24) **Q4** (170)

Q5 (4) **Q6** (2) **Q7** (55) **Q8** (6)

Q9 (32) **Q10** (2) **Q11** (3)

Q1

$$\lim_{x\to 1}\frac{\frac{1}{3}(5x+1)^{-2/3}5-\frac{1}{3}(x+5)^{-2/3}}{\frac{1}{2}(2x+3)^{-1/2}\cdot 2-\frac{1}{2}(x+4)^{-1/2}}$$

$$=\frac{8}{3}\frac{\sqrt{5}}{6^{2/3}}\begin{array}{l} m=8\\ n=3\end{array}$$

$$8m+12n=100$$

Q2

$\lim_{x\to 0}\frac{f(x)}{x^3}$

Using L Hopital Rule.

$\lim_{x\to 0}\frac{f'(x)}{3x^2}=\lim_{x\to 0}\frac{x+\sin(1-e^x)}{3x^2}$ (Again L Hopital)

Using L.H. Rule

$$=\lim_{x\to 0}\frac{-\left[\sin(1-e^x)\left(-e^x\right)\cdot e^x+\cos(1-e^x)\cdot e^x\right]}{6}$$

$$=-\frac{1}{6}$$

Q3

$$\lim_{t\to x}\frac{t^2f(x)-x^2f(t)}{t-x}=1$$

$$\lim_{t\to x}\frac{2t.\,f(x)-x^2f'(x)}{1}=1$$

$$2x\cdot f(x)-x2f'(x)=1$$

$$\frac{dy}{dx}-\frac{2}{x}\cdot y=\frac{-1}{x^2}$$

$$\text{I.f. }=e^{\int-\frac{2}{x}dx}=\frac{1}{x^2}$$

$$\therefore\frac{y}{x^2}=\int-\frac{1}{x^4}dx+C$$

$$\frac{y}{x^2}=\frac{1}{3x^3}+C$$

Put $f(1)=1$

$C=\frac{2}{3}$

$y=\frac{1}{3x}+\frac{2x^2}{3}$

$y=\frac{2x^3+1}{3x}$

$f(2)=\frac{17}{6}$

$f(3)=\frac{55}{9}$

$2f(2)+3f(3)=\frac{17}{3}+\frac{55}{3}=\frac{72}{3}=24$

Q4

$2x^2+x-2=0 \begin{cases} \nearrow a \\ \searrow b \end{cases}$

$2x^2-x-2=0 \begin{cases} \nearrow \frac{1}{a} \\ \searrow \frac{1}{b} \end{cases}$

$$\lim_{x\to\frac{1}{a}} 16\cdot\frac{\left(1-\cos 2\left(x-\frac{1}{a}\right)\left(x-\frac{1}{b}\right)\right)}{4\left(x-\frac{1}{b}\right)^2}\times\frac{4\left(x-\frac{1}{b}\right)^2}{a^2\left(x-\frac{1}{a}\right)^2}$$

$$=16\times\frac{2}{a^2}\left(\frac{1}{a}-\frac{1}{b}\right)^2$$

$$=\frac{32}{a^2}\left(\frac{17}{4}\right)=\frac{17.8}{a^2}=\frac{17\times 8\times 16}{(-1+\sqrt{117})^2}$$

$$=\frac{136.16}{18.2\sqrt{7}}\times\frac{18+2\sqrt{7}}{18+2\sqrt{7}}$$

$$=\frac{136}{256}(18+2\sqrt{7})\cdot 16$$

$$=153+17\sqrt{17}=\alpha+\beta\sqrt{17}$$

$\alpha+\beta=153+17=170$

Q5

$f:(-\infty, \infty)-\{0\} \rightarrow R$

$f'(1)=\lim_{a \rightarrow \infty} a^2 f\left(\frac{1}{a}\right)$

$\lim_{a \rightarrow \infty} \frac{a(a+1)}{2} \tan^{-1}\left(\frac{1}{a}\right)+a^2-2 \ln (a)$

$\lim_{a \rightarrow \infty} a^2\left(\frac{\left(1+\frac{1}{a}\right)}{2} \tan^{-1}\left(\frac{1}{a}\right)+1-\frac{2}{a^2} \ln (a)\right)$

$f(x)=\frac{1}{2}(1+x) \tan^{-1}(x)+1-2x^2 \ln (x)$

$\mathrm{f}'(\mathrm{x})=\frac{1}{2}\left(\frac{1+\mathrm{x}}{1+\mathrm{x}^2}+\tan^{-1}(\mathrm{x})+4\mathrm{x} \ell(\mathrm{x})\right)+2\mathrm{x}$

$\mathrm{f}'(1)=\frac{1}{2}\left(1+\frac{\pi}{4}\right)+2$

$\mathrm{f}'(1)=\frac{5}{2}+\frac{\pi}{8}$

Q6

$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^{n-1}\left(r^2-r\right)(n-r)}{\sum_{r=1}^n r^3-\sum_{r=1}^n r^2}$$

$$\lim_{\mathrm{n} \rightarrow \infty} \frac{\sum_{\mathrm{r}=1}^{\mathrm{n}-1}\left(-\mathrm{r}^3+\mathrm{r}^2(\mathrm{n}+1)-\mathrm{nr}\right)}{\left(\frac{\mathrm{n}(\mathrm{n}+1)}{2}\right)^2-\frac{\mathrm{n}(\mathrm{n}+1)(2\mathrm{n}+1)}{6}}$$

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{((n-1) n)}{2}\right)^2+\frac{(n+1)(n-1) n(2n-1)}{6}-\frac{n^2(n-1)}{2}}{\frac{n(n+1)}{2}\left(\frac{n(n+1)}{2}-\frac{2n+1}{3}\right)}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n(n-1)}{2}\left(\frac{-n(n-1)}{2}+\frac{(n+1)(2n-1)}{3}-n\right)}{\frac{n(n+1)}{2} \frac{3n^2+3n-4n-2}{6}}$$

$$\lim_{n \rightarrow \infty} \frac{(n-1)\left(-3n^2+3n+2\left(2n^2+n-1\right)-6\right)}{(n+1)\left(3n^2-n-2\right)}$$

$$\lim_{n \rightarrow \infty} \frac{(n-1)\left(n^2+5n-8\right)}{(n+1)\left(3n^2-n-2\right)}=\frac{1}{3}$$

Q7

$\lim_{x\to 0} 2\left(\frac{1-\left(1-\frac{x^2}{2!}\right)\left(1-\frac{4x^2}{2!}\right)\left(1-\frac{9x^2}{2!}\right)\ldots\left(1-\frac{100x^2}{2!}\right)}{x^2}\right)$

By expansion

$$\lim_{x\to 0}\frac{2\left(1-\left(1-\frac{x^2}{2}\right)\right)\left(1-\frac{1}{2}\cdot\frac{4x^2}{2}\right)\left(1-\frac{1}{3}\cdot\frac{9x^2}{2}\right)\ldots\ldots\left(1-\frac{1}{10}\cdot\frac{100x^2}{2}\right)}{x^2}\Bigg)$$

$$\lim_{x\to 0} 2\left(\frac{1-\left(1-\frac{x^2}{2}\right)\left(1-\frac{2x^2}{2}\right)\left(1-\frac{3x^2}{2}\right)\ldots\ldots\left(1-\frac{10x^2}{2}\right)\Big)}{x^2}\right)$$

$$\lim_{x\to 0}\frac{2\left(1-1+x^2\left(\frac{1}{2}+\frac{2}{2}+\frac{3}{2}+\ldots\ldots+\frac{10}{2}\right)\right)}{x^2}$$

$$2\left(\frac{1}{2}+\frac{2}{2}+\frac{3}{2}+\ldots\ldots+\frac{10}{2}\right)$$

$$1+2+\ldots\ldots+10=\frac{10\times 11}{2}=55$$

Q8

$$\alpha=\lim_{x\to 0^+} e^{\sqrt{x}}\frac{\left(e^{\sqrt{\tan x}-\sqrt{x}}-1\right)}{\sqrt{\tan x}-\sqrt{x}}$$

$=1$

$\beta=\lim_{x\to 0}(1+\sin x)^{\frac{1}{2}\cot x}$

$=e^{1/2}$

$x^2-(1+\sqrt{e})+\sqrt{e}=0$

$ax^2+bx-\sqrt{e}=0$

On comparing

$a=-1, b=\sqrt{e}+1$

$12\ln(a+b)=12\times\frac{1}{2}=6$

Q9

$$\sum_{r=1}^{\infty}\frac{n}{\sqrt{n^4+r^4}}-\frac{2nr^2}{\left(n^2+r^2\right)\sqrt{n^4+r^4}}$$

$$\sum_{r=1}^{\infty}\frac{\frac{1}{n}}{\sqrt{1+\left(\frac{r}{n}\right)^4}}-\frac{2\left(\frac{1}{n}\right)\left(\frac{r}{n}\right)^2}{\left(1+\left(\frac{r}{n}\right)^2\right)\sqrt{1+\left(\frac{r}{n}\right)^4}}$$

$$\Rightarrow\int_0^1\frac{dx}{\sqrt{1+x^4}}-\frac{2x^2dx}{\left(1+x^2\right)\sqrt{1+x^4}}$$

$$\Rightarrow\int_0^1\frac{1-x^2}{\left(1+x^2\right)\sqrt{1+x^4}}dx$$

$$\Rightarrow\int_0^1\frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}dx$$

$$\Rightarrow-\int_0^1\frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx$$

$$x+\frac{1}{x}=t\Rightarrow 1-\frac{1}{x^2}dx=dt$$

$$\Rightarrow-\int_{\infty}^{2}\frac{dt}{t\sqrt{t^2-2}}$$

$$\Rightarrow-\int_{\infty}^{2}\frac{tdt}{t^2\sqrt{t^2-2}}$$

take $t^2-2=\alpha^2$

$tdt=\alpha d\alpha$

$$\Rightarrow-\int_{\infty}^{\sqrt{2}}\frac{\alpha d\alpha}{\left(\alpha^2+2\right)\alpha}$$

$$\Rightarrow-\int_{\infty}^{\sqrt{2}}\frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow\left.\frac{-1}{\sqrt{2}}\tan^{-1}\frac{\alpha}{\sqrt{2}}\right]_{\infty}^{\sqrt{2}}$$

$$\Rightarrow\frac{-1}{\sqrt{2}}\left\{\tan^{-1}1\right\}+\frac{1}{\sqrt{2}}\tan^{-1}\infty$$

$$\Rightarrow\frac{1}{\sqrt{2}}\left\{\frac{\pi}{2}-\frac{\pi}{4}\right\}$$

$$\Rightarrow\frac{\pi}{4\sqrt{2}}=\frac{\pi}{K}$$

So $K=4\sqrt{2}$

$K^2=32$

Q10

$$\lim_{x \to \frac{\pi}{2}} \frac{0-\{\sin(2x)+\cos(x)\} \cdot 3x^2}{2\left(x-\frac{\pi}{2}\right)}$$

$$=\lim_{x \to \frac{\pi}{2}} \frac{-\{2 \sin x \cos x+\cos x\} 3x^2}{2\left(x-\frac{\pi}{2}\right)}$$

$$=\lim_{x \to \frac{\pi}{2}}\left\{\frac{2 \sin x \sin \left(\frac{\pi}{2}-x\right)}{2\left(x-\frac{\pi}{2}\right)}+\frac{\sin \left(\frac{\pi}{2}-x\right)}{2\left(\frac{\pi}{2}-x\right)}\right\} 3x^2$$

$$=\left(1(1)+\frac{1}{2}\right) 3\left(\frac{\pi}{2}\right)^2$$

$$=\frac{9\pi^2}{8}$$

Q11

$$\operatorname{Lim}_{x \to 0} \frac{e-e^{\frac{1}{2x} \ln(1+2x)}}{x}$$

$$=\operatorname{Lim}_{x \to 0}(-e) \frac{\left(e^{\frac{\ln(1+2x)}{2x}-1}-1\right)}{x}$$

$$=\operatorname{Lim}_{x \to 0}(-e) \frac{\ln(1+2x)-2x}{2x^2}$$

$$=(-e) \times(-1) \frac{4}{2 \times 2}=e$$